DeepFold3

Kernel Optimization Report

GPU: NVIDIA GeForce RTX 5090 | dtype: bfloat16 | 2026-03-17

Optimization Strategy

```
KEY RESULTS (fwd+bwd, best shape)

  torch.compile                  3.41x   GridSelfAttention (N=640 C=128 H=4)
  EvoAttn+PairBias               3.20x   GridColAttn (N=640 C=128 H=4)
  torch.compile                  2.36x   MSAAttention (N=768 C_msa=64 C_pair=128 H=8 S=16)

OPTIMIZATION STACK (adapted from NVIDIA cuEquivariance)

  ATTENTION (104 modules)
    Fused EvoformerAttention — Triton FlashAttention + pair bias
    Fused Pair Bias — LN + tl.dot GEMM(128→4) + permute + mask

  TRIANGLE MULTIPLY (96 modules)
    L3: Fused Dual GEMM — sigmoid(x@W1) * (x2@W2) * mask
    L2: Fused LN+Linear for post-einsum norm_center + out_proj
    L1: Fused LN + packed GEMM + sigmoid for pre-einsum gate

  SWIGLU (TransitionBlock / AdaTransitionBlock)
    Multi-row kernel (128 rows/block) + in-place backward
    Pair (M=N²): 1.5-2.1x faster, 2-6x less memory
    Single (M=N): neutral speed, still saves memory

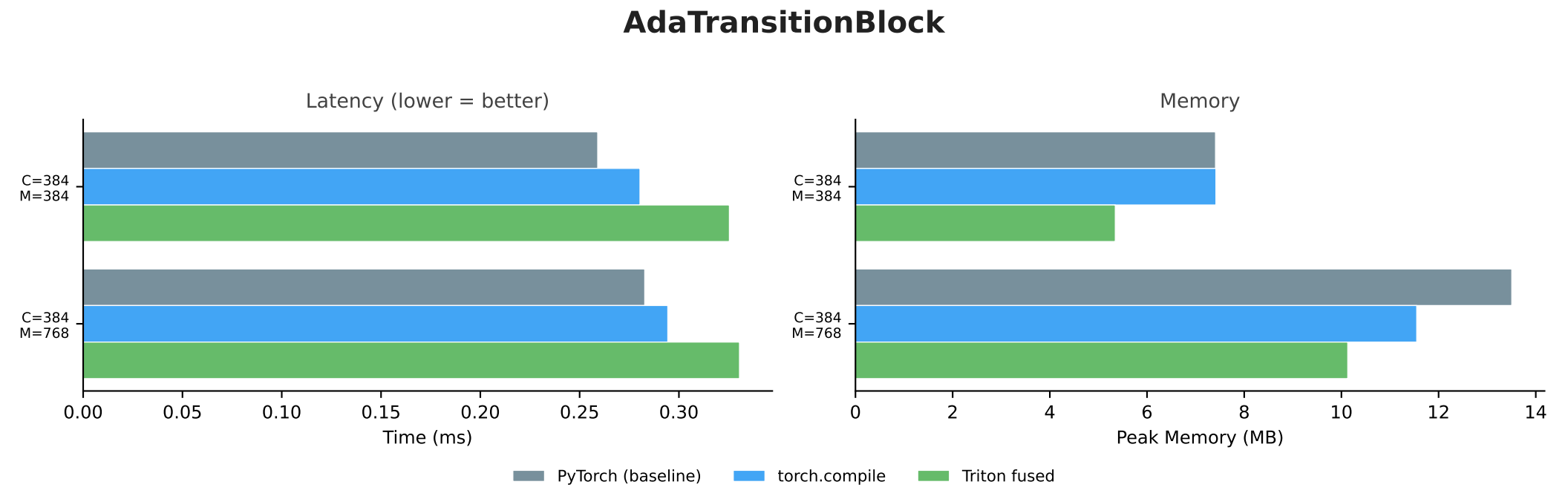
  TRANSITIONS (context-sensitive)
    Pair (M=N², C=128): Triton fused → 1.2-1.3x
    Single (M=N, C=384): PyTorch baseline faster (M too small)

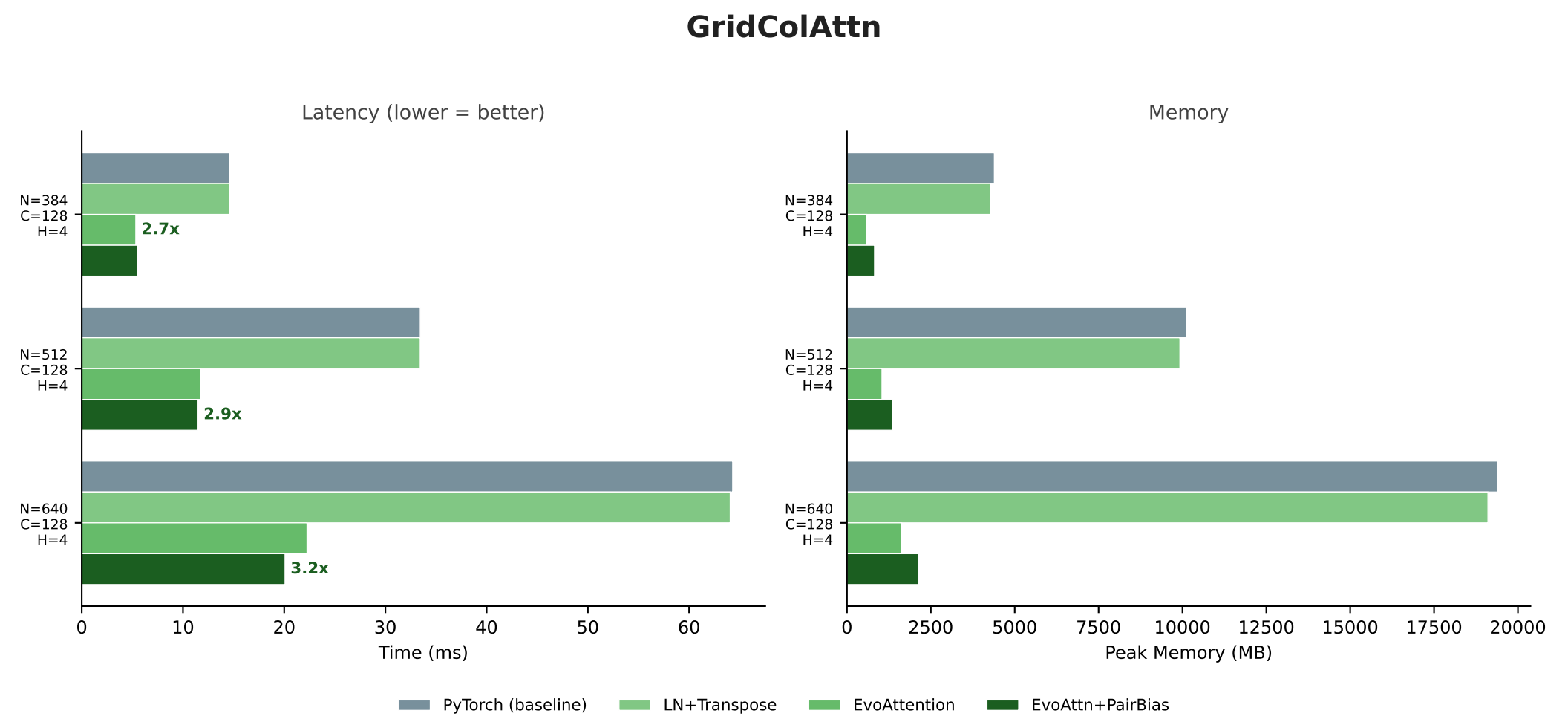
  INFRASTRUCTURE
    Persistent autotuning cache with GPU fingerprinting
    Precision control: DEFAULT / TF32 / TF32x3 / IEEE
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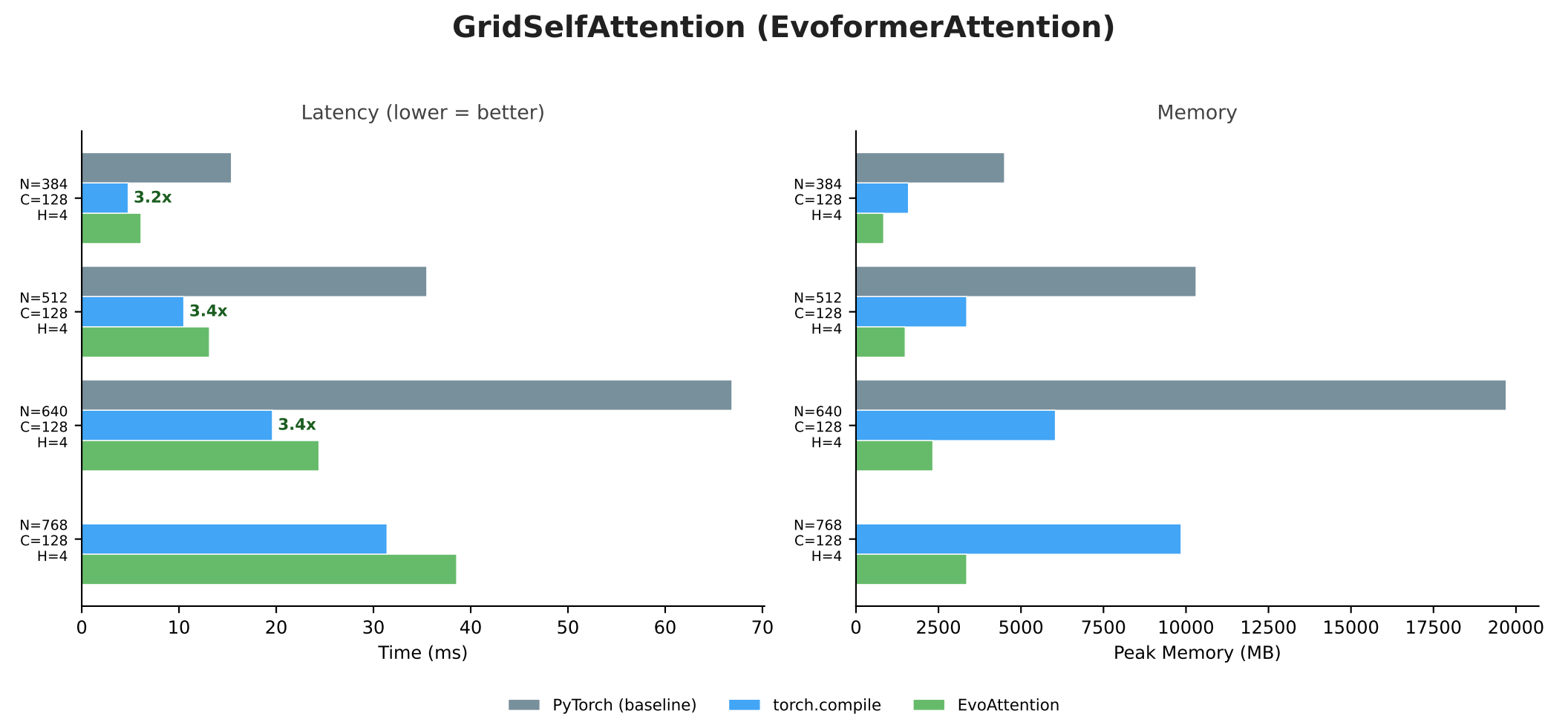
Numerical Verification

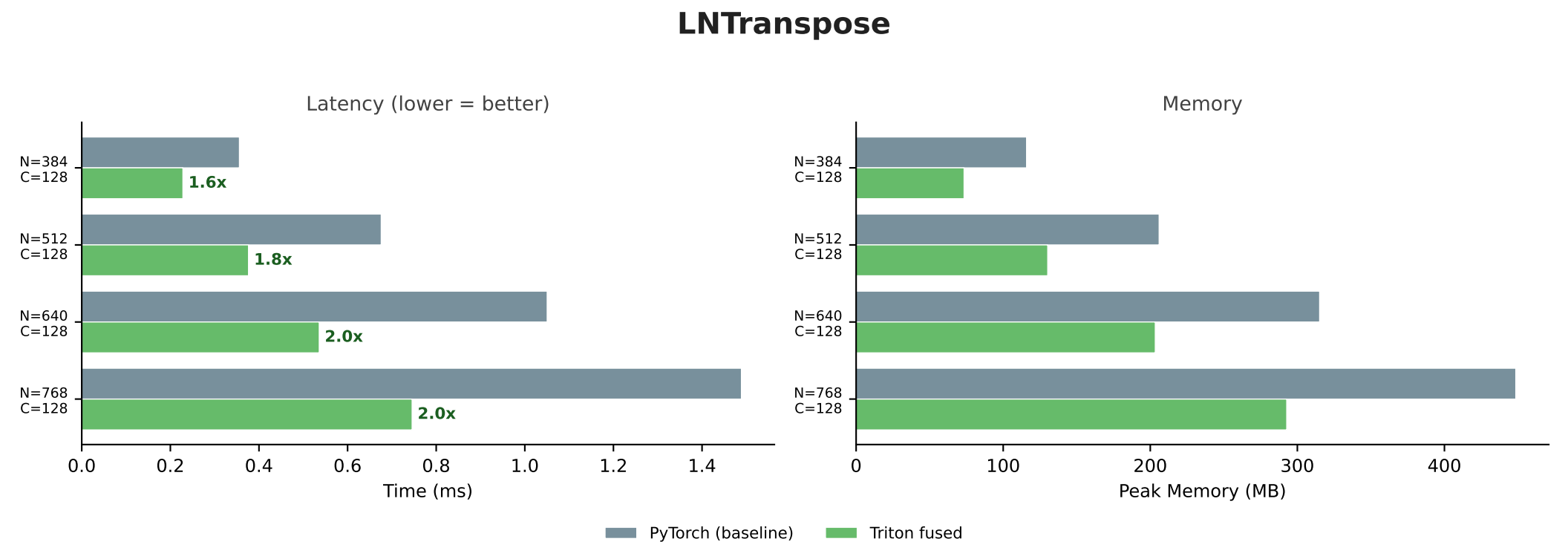
All fused kernels verified against PyTorch baseline (bf16, atol=5e-2, rtol=5e-2)

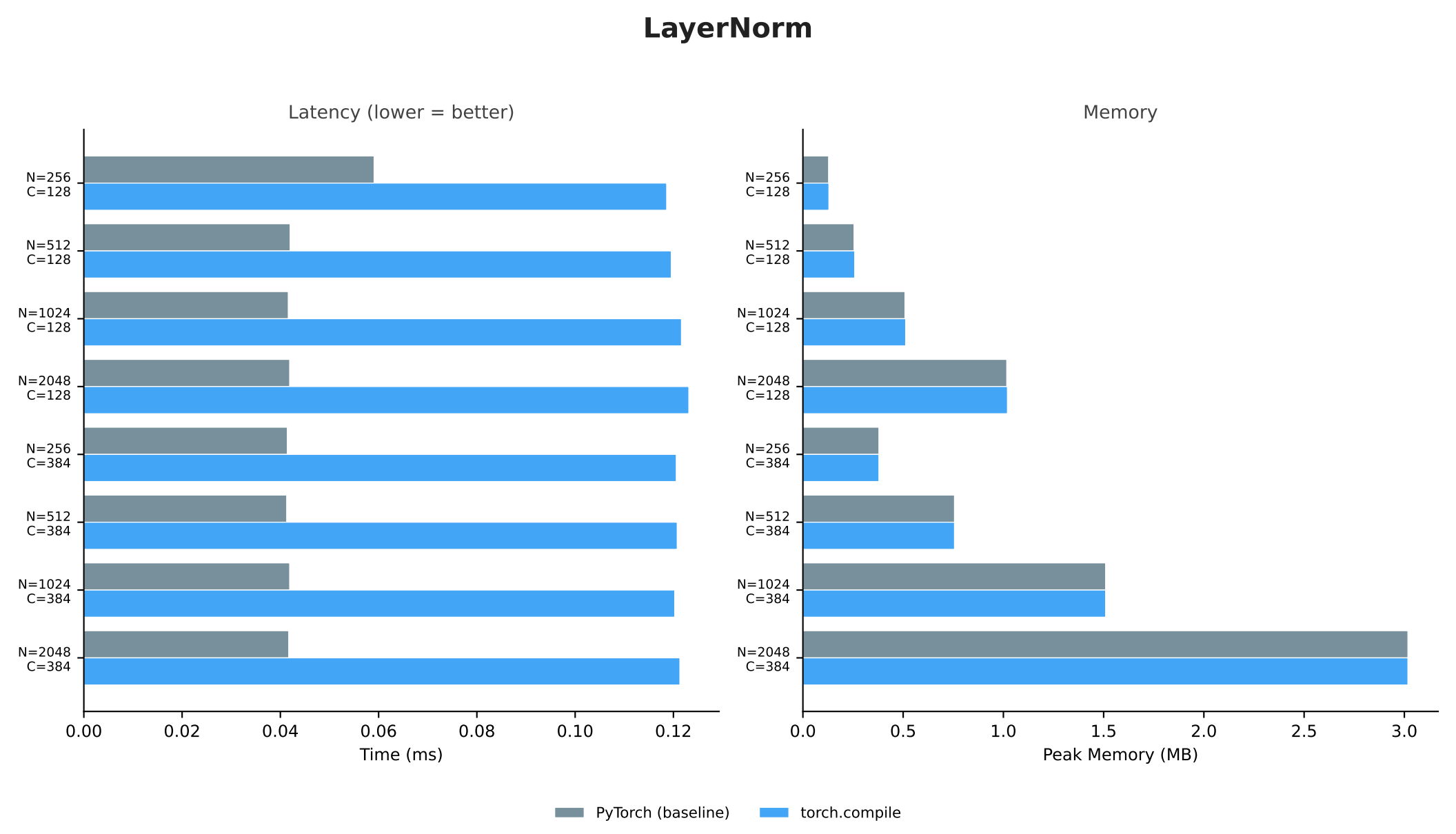
Kernel	Forward	Fwd Max Diff	Backward	Bwd Max Diff
SwiGLU (Triton)	✔ PASS	0.000000	✔ PASS	0.062500
LayerNormLinear (Triton)	✔ PASS	0.000000	✔ PASS	0.000000
TransitionBlock (Triton)	✔ PASS	0.000000	✔ PASS	0.003906
EvoformerAttention (Triton)	✔ PASS	0.007812	✔ PASS	0.017555
SingleAttention + EvoAttention (Triton)	✔ PASS	0.000000	✔ PASS	0.000000
GridSelfAttention + EvoAttention (Triton)	✔ PASS	0.000977	✔ PASS	0.002441
AdaLayerNormLinear (Triton)	✔ PASS	0.000000	✔ PASS	0.000000
AdaTransitionBlock (Triton)	✔ PASS	0.000000	✔ PASS	0.000244
TriangleMulGate (Triton)	✔ PASS	0.000000	✔ PASS	0.015625
TriangleMul E2E (fused gate)	✔ PASS	0.015625	✔ PASS	0.023438
LayerNormTranspose (Triton)	✔ PASS	0.007812	✔ PASS	0.000000
Fused Dual GEMM (Triton)	✔ PASS	0.031250	✔ PASS	0.031250
TriangleMul center proj (Triton)	✔ PASS	0.015625	✔ PASS	0.023438
GridSelfAttention LN+transpose (Triton)	✔ PASS	0.000000	✔ PASS	0.000977
	✔ ALL PASSED			

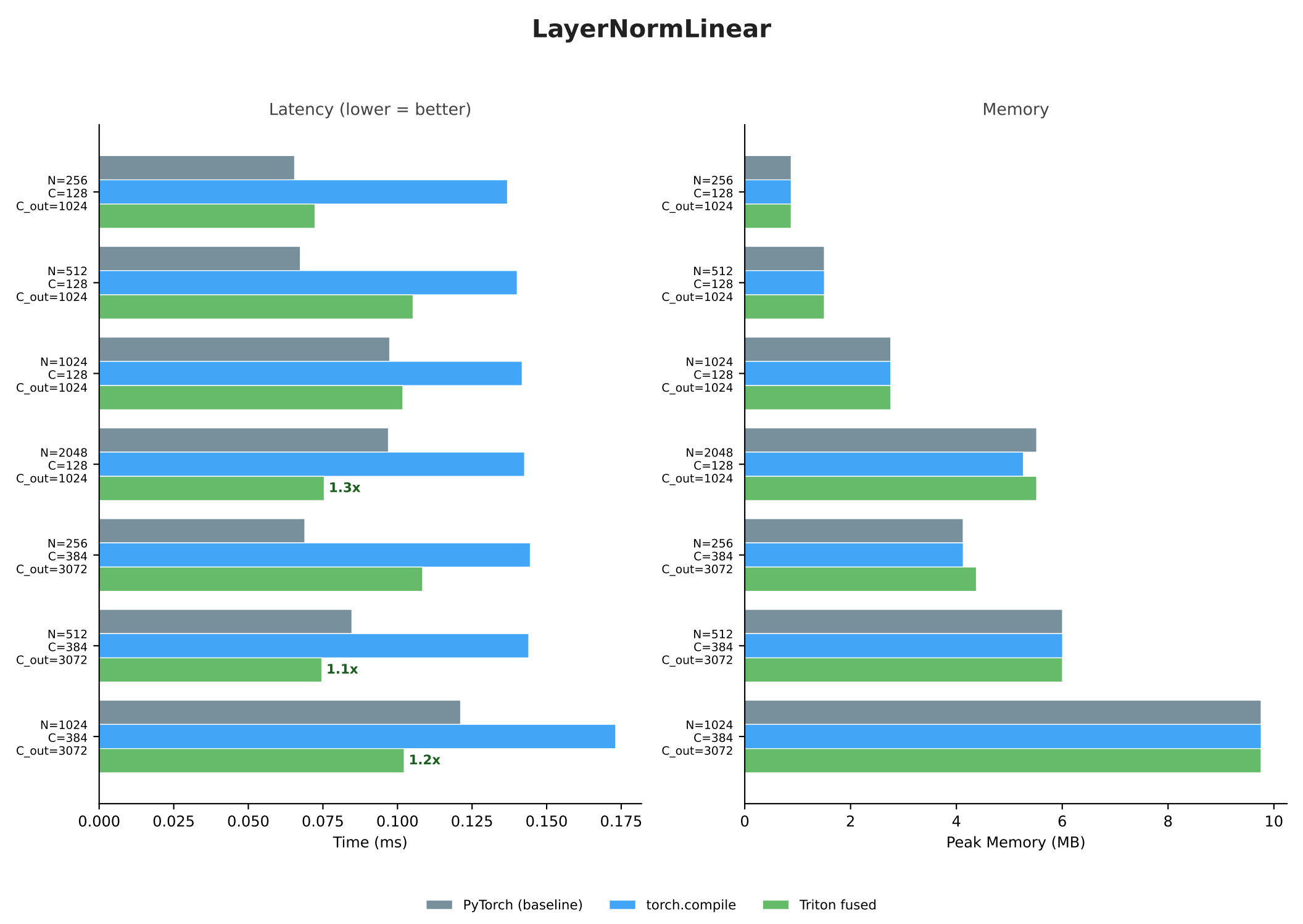

AdaTransitionBlock
Latency (lower = better)
C=384
M=384
C=384
M=768
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Time (ms)
Memory
C=384
M=384
C=384
M=768
0
2
4
6
8
10
12
14
Peak Memory (MB)
PyTorch (baseline)
torch.compile
Triton fused

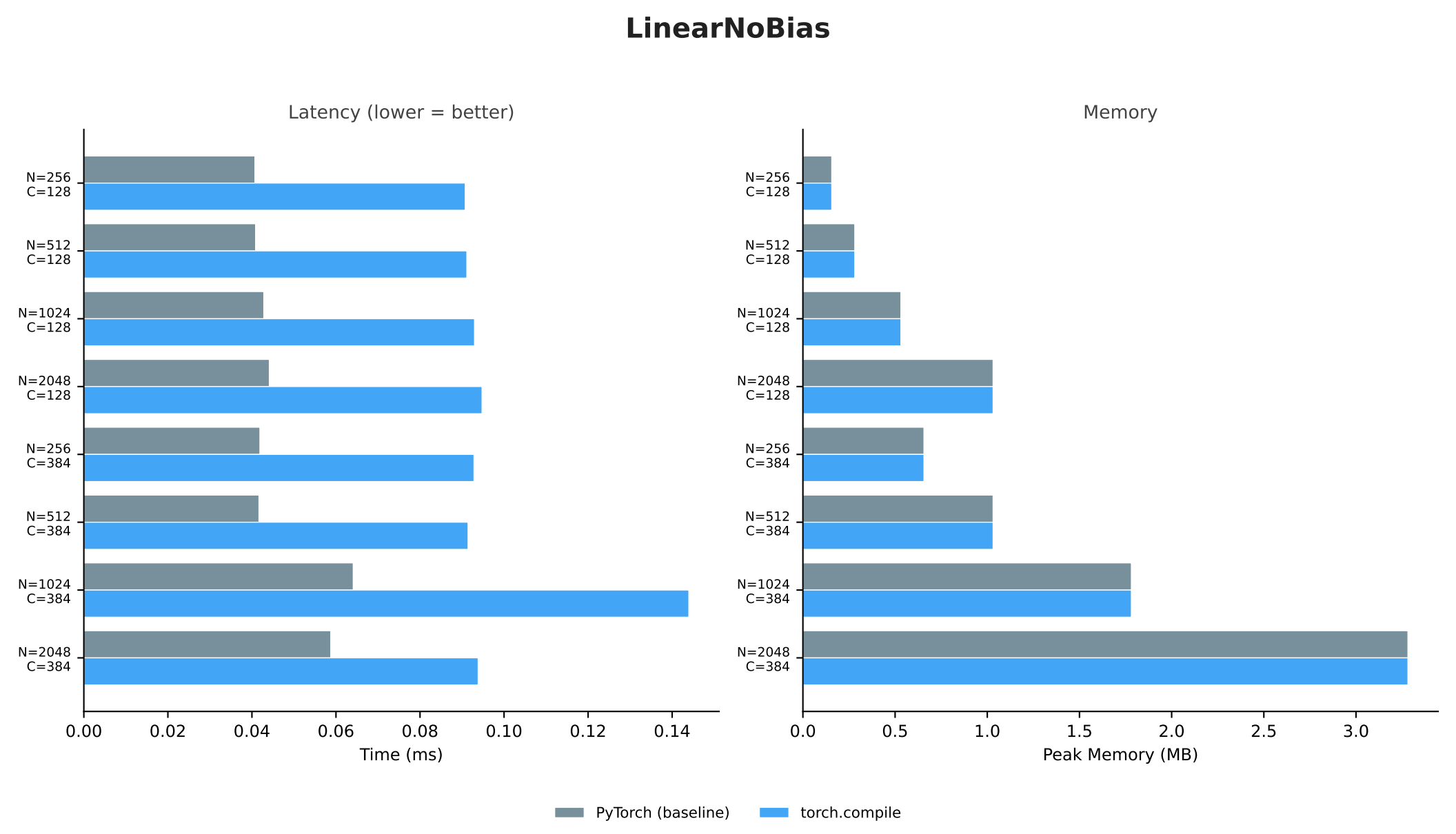

GridColAttn
Latency (lower = better)
Memory
N=384
C=128
H=4
N=512
C=128
H=4
N=640
C=128
H=4
2.7x
2.9x
3.2x
0
10
20
30
40
50
60
Time (ms)
0
2500
5000
7500
10000
12500
15000
17500
20000
Peak Memory (MB)
PyTorch (baseline)
LN+Transpose
EvoAttention
EvoAttn+PairBias

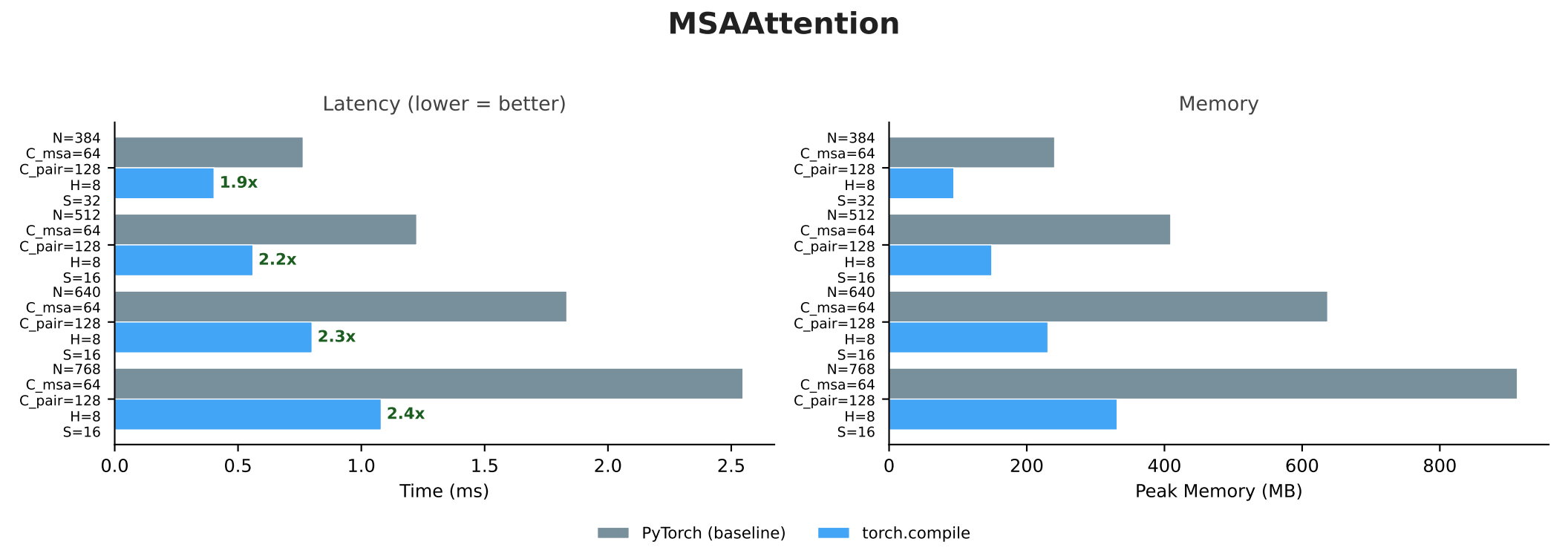

GridSelfAttention (EvoformerAttention)
Latency (lower = better)
Memory
N=384
C=128
H=4
N=512
C=128
H=4
N=640
C=128
H=4
N=768
C=128
H=4
3.2x
3.4x
3.4x
0
10
20
30
40
50
60
70
Time (ms)
0
2500
5000
7500
10000
12500
15000
17500
20000
Peak Memory (MB)
PyTorch (baseline)
torch.compile
EvoAttention

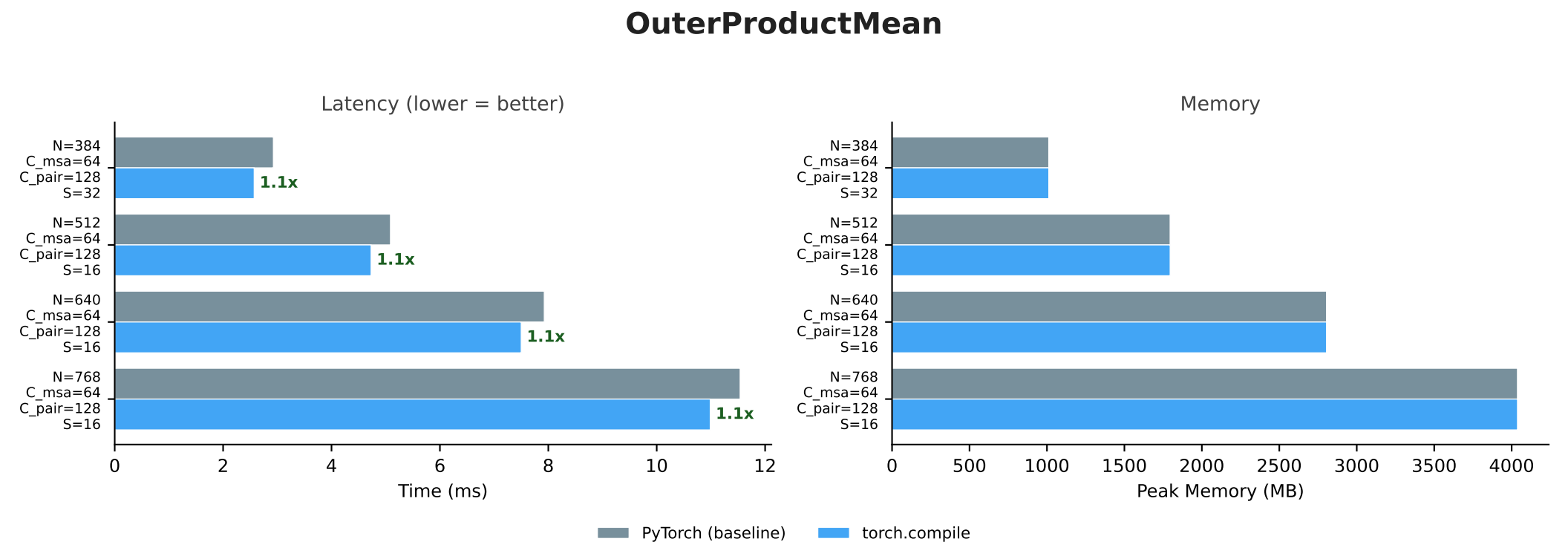

LNTranspose
Latency (lower = better)
N=384
C=128
1.6x
N=512
C=128
1.8x
N=640
C=128
2.0x
N=768
C=128
2.0x
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time (ms)
Memory
N=384
C=128
N=512
C=128
N=640
C=128
N=768
C=128
0
100
200
300
400
Peak Memory (MB)
PyTorch (baseline)
Triton fused

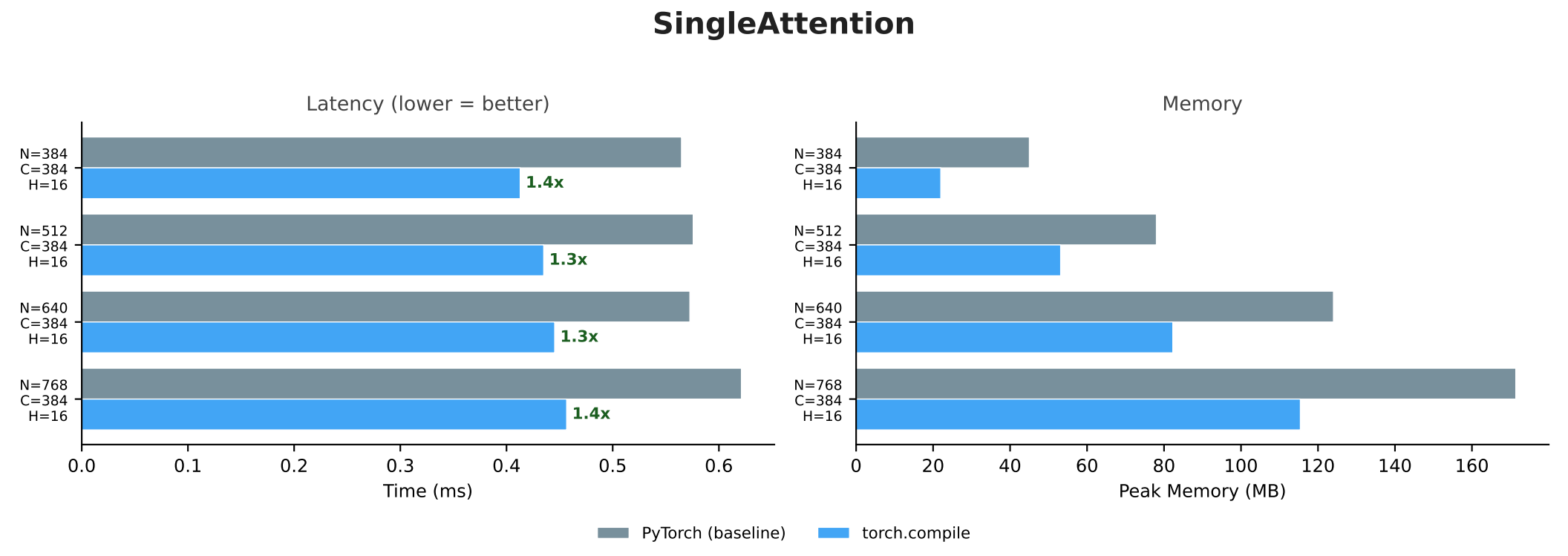

LayerNorm
Latency (lower = better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Time (ms)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Peak Memory (MB)
PyTorch (baseline)
torch.compile


LayerNormLinear
Latency (lower = better)
Memory
N=256 C=128 C_out=1024
N=512 C=128 C_out=1024
N=1024 C=128 C_out=1024
N=2048 C=128 C_out=1024
N=256 C=384 C_out=3072
N=512 C=384 C_out=3072
N=1024 C=384 C_out=3072
1.3x
1.1x
1.2x
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Time (ms)
0
2
4
6
8
10
Peak Memory (MB)
PyTorch (baseline)
torch.compile
Triton fused

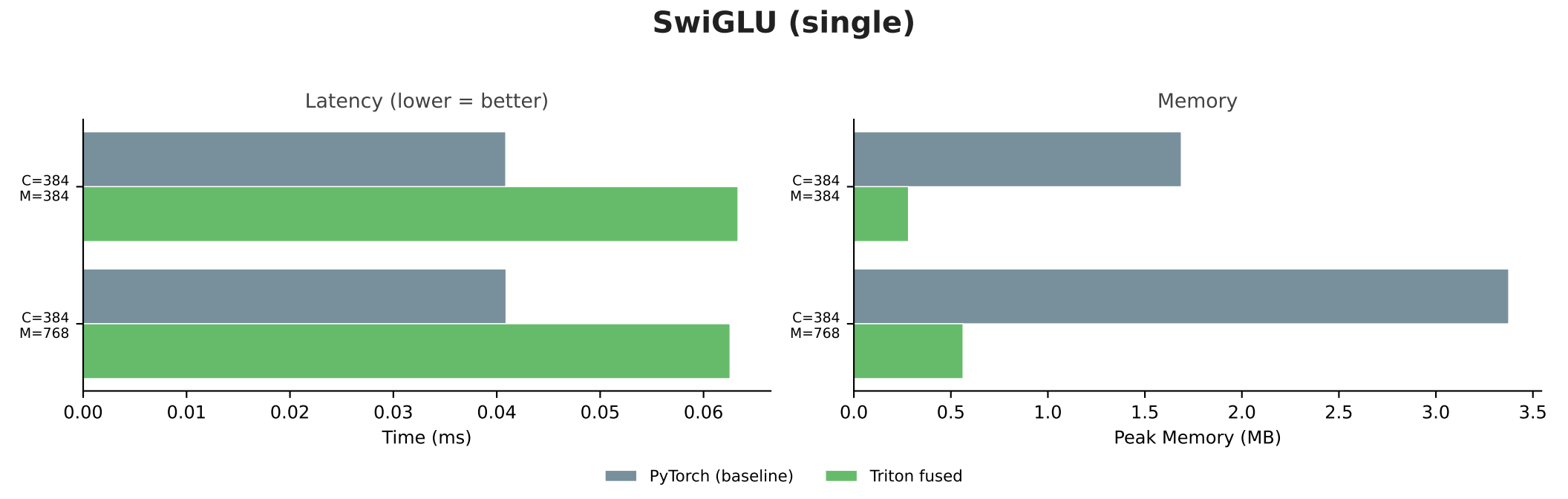

LinearNoBias
Latency (lower = better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Time (ms)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Peak Memory (MB)
PyTorch (baseline)
torch.compile

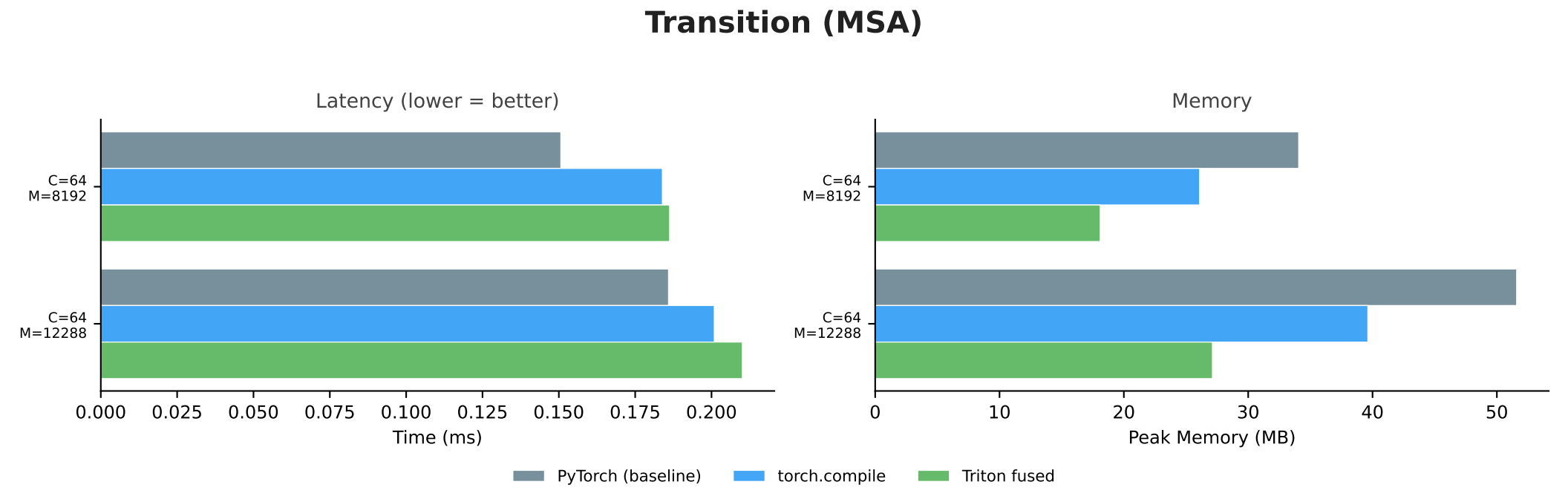

MSAAttention
Latency (lower = better)
N=384
C_msa=64
C_pair=128
H=8
S=32
1.9x
N=512
C_msa=64
C_pair=128
H=8
S=16
2.2x
N=640
C_msa=64
C_pair=128
H=8
S=16
2.3x
N=768
C_msa=64
C_pair=128
H=8
S=16
2.4x
0.0
0.5
1.0
1.5
2.0
2.5
Time (ms)
Memory
0
200
400
600
800
Peak Memory (MB)
PyTorch (baseline)
torch.compile

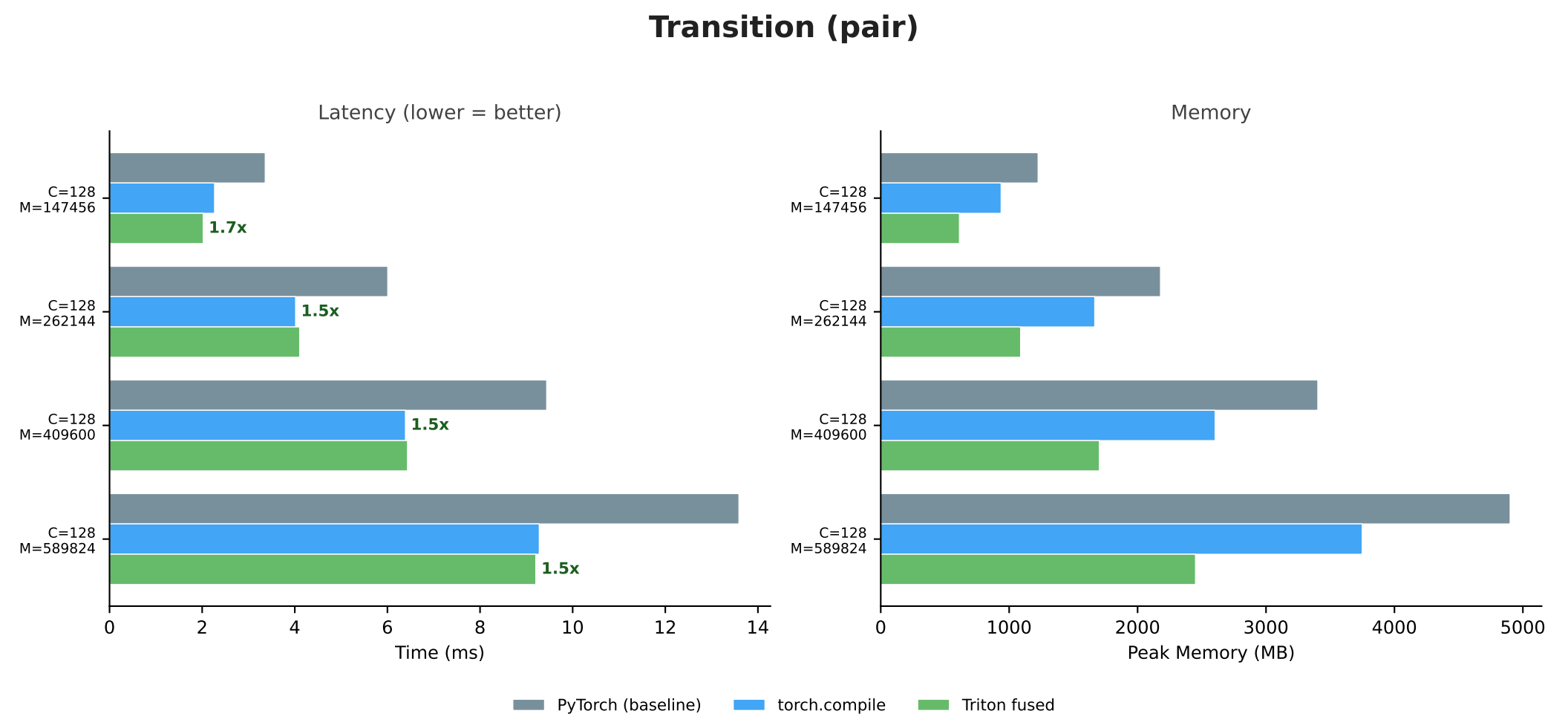

OuterProductMean
Latency (lower = better)
N=384
C_msa=64
C_pair=128
S=32
1.1x
N=512
C_msa=64
C_pair=128
S=16
1.1x
N=640
C_msa=64
C_pair=128
S=16
1.1x
N=768
C_msa=64
C_pair=128
S=16
1.1x
0
2
4
6
8
10
12
Time (ms)
Memory
N=384
C_msa=64
C_pair=128
S=32
N=512
C_msa=64
C_pair=128
S=16
N=640
C_msa=64
C_pair=128
S=16
N=768
C_msa=64
C_pair=128
S=16
0
500
1000
1500
2000
2500
3000
3500
4000
Peak Memory (MB)
PyTorch (baseline)
torch.compile

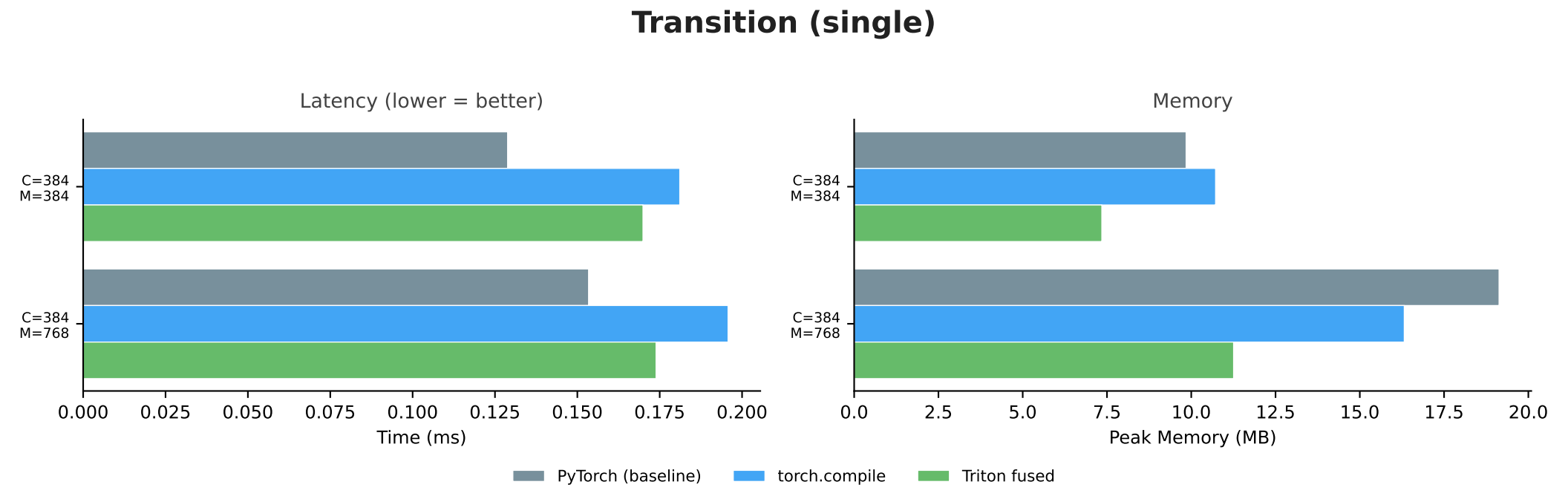

SingleAttention
Latency (lower = better)
N=384
C=384
H=16
1.4x
N=512
C=384
H=16
1.3x
N=640
C=384
H=16
1.3x
N=768
C=384
H=16
1.4x
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Time (ms)
Memory
N=384
C=384
H=16
N=512
C=384
H=16
N=640
C=384
H=16
N=768
C=384
H=16
0
20
40
60
80
100
120
140
160
Peak Memory (MB)
PyTorch (baseline)
torch.compile

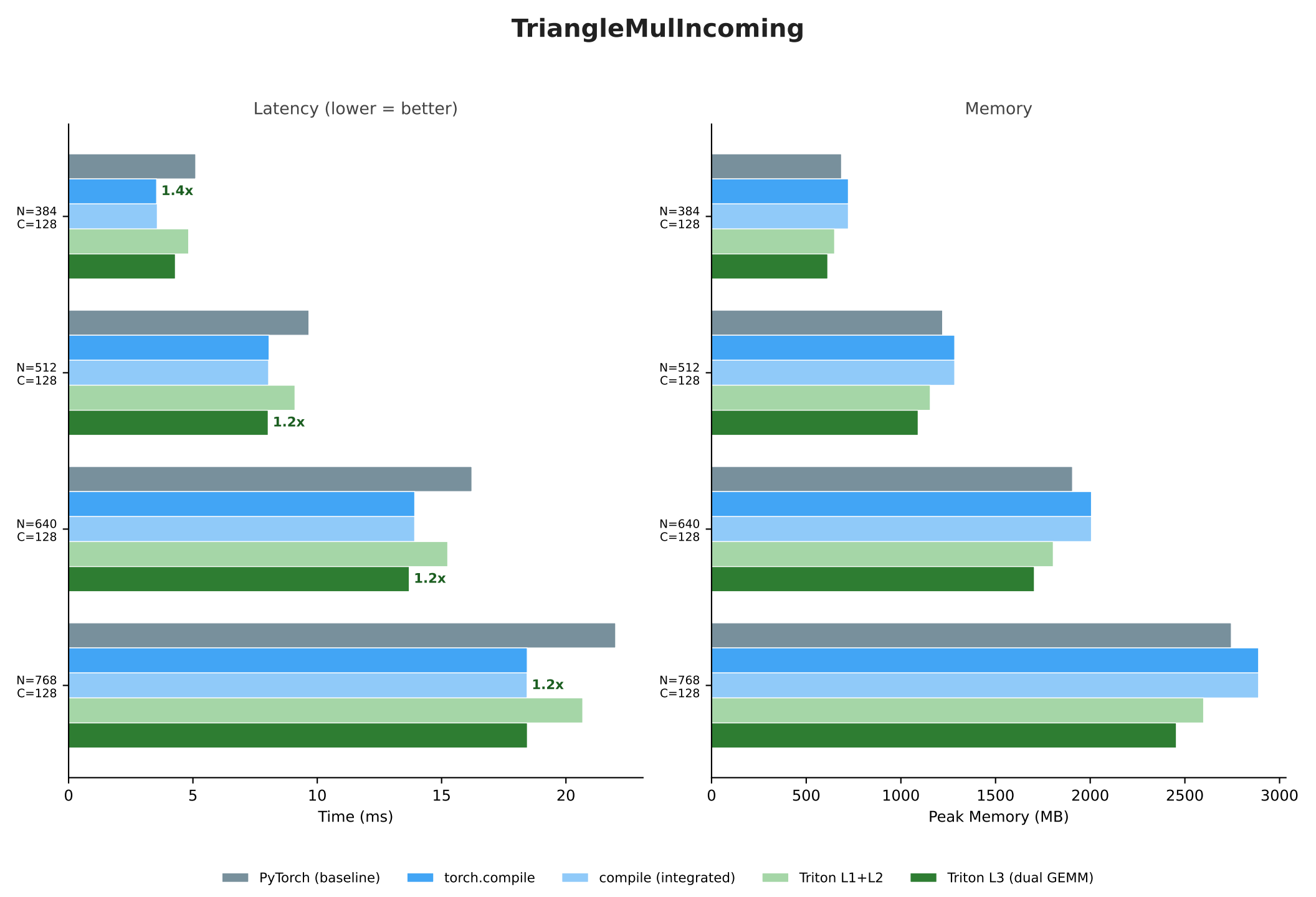

SwiGLU (pair)
Latency (lower = better)
C=128
M=147456
1.9x
C=128
M=262144
1.9x
C=128
M=409600
2.1x
C=128
M=589824
2.1x
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Time (ms)
Memory
C=128
M=147456
C=128
M=262144
C=128
M=409600
C=128
M=589824
0
200
400
600
800
Peak Memory (MB)
PyTorch (baseline)
Triton fused

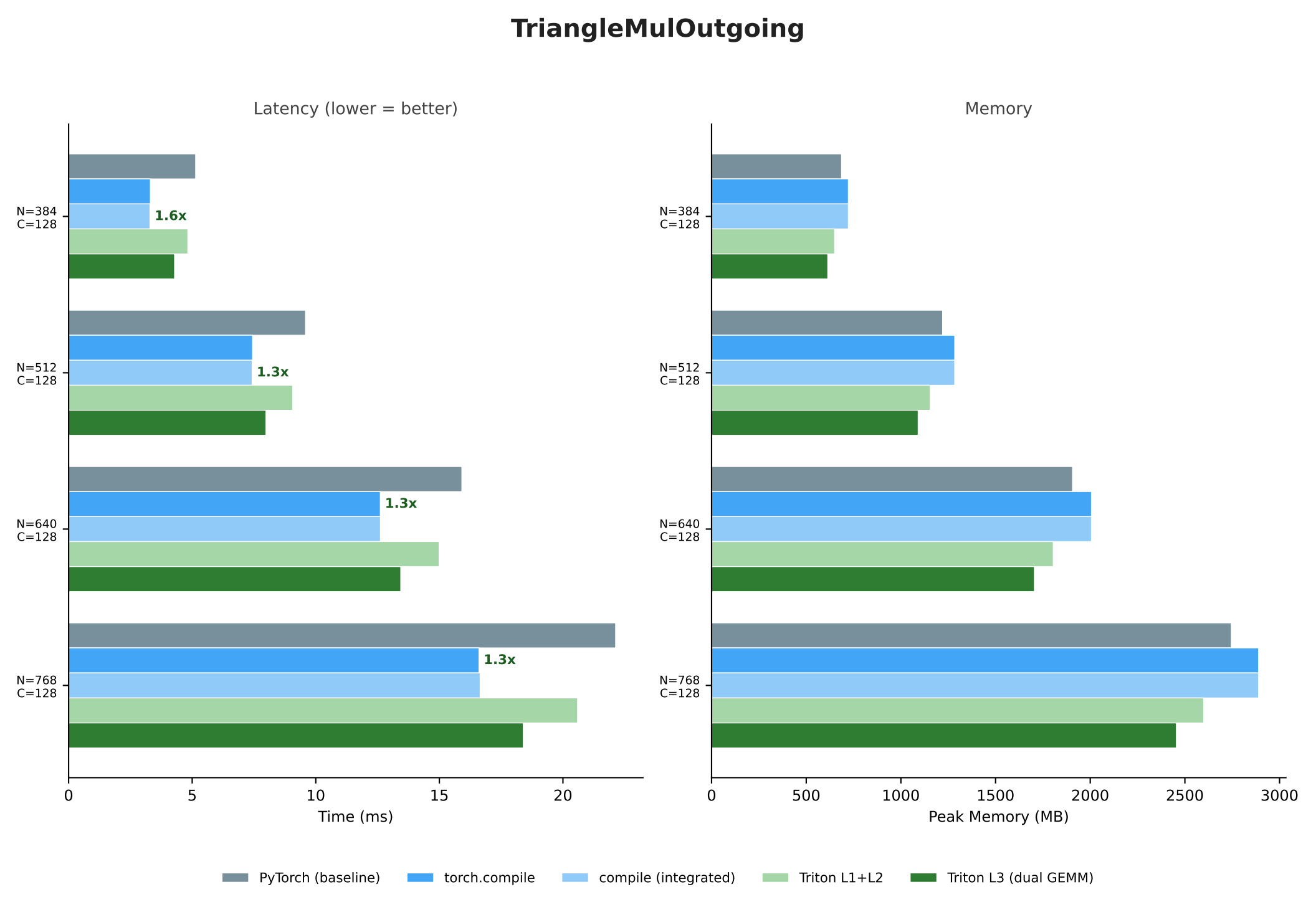

SwiGLU (single)
Latency (lower = better)
C=384
M=384
C=384
M=768
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Time (ms)
Memory
C=384
M=384
C=384
M=768
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Peak Memory (MB)
PyTorch (baseline)
Triton fused


Transition (MSA)
Latency (lower = better)
C=64
M=8192
C=64
M=12288
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Time (ms)
Memory
C=64
M=8192
C=64
M=12288
0
10
20
30
40
50
Peak Memory (MB)
PyTorch (baseline)
torch.compile
Triton fused


Transition (pair)
Latency (lower = better)
Memory
C=128
M=147456
C=128
M=262144
C=128
M=409600
C=128
M=589824
1.7x
1.5x
1.5x
1.5x
0
2
4
6
8
10
12
14
Time (ms)
0
1000
2000
3000
4000
5000
Peak Memory (MB)
PyTorch (baseline)
torch.compile
Triton fused


Transition (single)
Latency (lower = better)
C=384
M=384
C=384
M=768
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Time (ms)
Memory
C=384
M=384
C=384
M=768
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Peak Memory (MB)
PyTorch (baseline)
torch.compile
Triton fused


TriangleMulIncoming
Latency (lower = better)
Memory
N=384
C=128
N=512
C=128
N=640
C=128
N=768
C=128
1.4x
1.2x
1.2x
1.2x
0
5
10
15
20
Time (ms)
0
500
1000
1500
2000
2500
3000
Peak Memory (MB)
PyTorch (baseline)
torch.compile
compile (integrated)
Triton L1+L2
Triton L3 (dual GEMM)


TriangleMulOutgoing
Latency (lower = better)
Memory
N=384
C=128
N=512
C=128
N=640
C=128
N=768
C=128
1.6x
1.3x
1.3x
1.3x
0
5
10
15
20
Time (ms)
0
500
1000
1500
2000
2500
3000
Peak Memory (MB)
PyTorch (baseline)
torch.compile
compile (integrated)
Triton L1+L2
Triton L3 (dual GEMM)

Recommendation Summary

Operation	Shape	Recommended	Time (ms)	Speedup
AdaTransitionBlock	C=384 / M=384	PyTorch (baseline)	0.26	—
AdaTransitionBlock	C=384 / M=768	PyTorch (baseline)	0.28	—
GridColAttn	N=384 / C=128 / H=4	EvoAttention	5.34	2.7x
GridColAttn	N=512 / C=128 / H=4	EvoAttn+PairBias	11.49	2.9x
GridColAttn	N=640 / C=128 / H=4	EvoAttn+PairBias	20.08	3.2x
GridSelfAttention	N=384 / C=128 / H=4	torch.compile	4.77	3.2x
GridSelfAttention	N=512 / C=128 / H=4	torch.compile	10.49	3.4x
GridSelfAttention	N=640 / C=128 / H=4	torch.compile	19.61	3.4x
GridSelfAttention	N=768 / C=128 / H=4	torch.compile	31.39	—
LNTranspose	N=384 / C=128	Triton fused	0.23	1.6x
LNTranspose	N=512 / C=128	Triton fused	0.38	1.8x
LNTranspose	N=640 / C=128	Triton fused	0.54	2.0x
LNTranspose	N=768 / C=128	Triton fused	0.75	2.0x
LayerNorm	N=256 / C=128	PyTorch (baseline)	0.06	—
LayerNorm	N=512 / C=128	PyTorch (baseline)	0.04	—
LayerNorm	N=1024 / C=128	PyTorch (baseline)	0.04	—
LayerNorm	N=2048 / C=128	PyTorch (baseline)	0.04	—
LayerNorm	N=256 / C=384	PyTorch (baseline)	0.04	—
LayerNorm	N=512 / C=384	PyTorch (baseline)	0.04	—
LayerNorm	N=1024 / C=384	PyTorch (baseline)	0.04	—
LayerNorm	N=2048 / C=384	PyTorch (baseline)	0.04	—
LayerNormLinear	N=256 / C=128 / C_out=1024	PyTorch (baseline)	0.07	—
LayerNormLinear	N=512 / C=128 / C_out=1024	PyTorch (baseline)	0.07	—
LayerNormLinear	N=1024 / C=128 / C_out=1024	PyTorch (baseline)	0.10	—
LayerNormLinear	N=2048 / C=128 / C_out=1024	Triton fused	0.08	1.3x
LayerNormLinear	N=256 / C=384 / C_out=3072	PyTorch (baseline)	0.07	—
LayerNormLinear	N=512 / C=384 / C_out=3072	Triton fused	0.07	1.1x
LayerNormLinear	N=1024 / C=384 / C_out=3072	Triton fused	0.10	1.2x

Operation	Shape	Recommended	Time (ms)	Speedup
LinearNoBias	N=256 / C=128	PyTorch (baseline)	0.04	—
LinearNoBias	N=512 / C=128	PyTorch (baseline)	0.04	—
LinearNoBias	N=1024 / C=128	PyTorch (baseline)	0.04	—
LinearNoBias	N=2048 / C=128	PyTorch (baseline)	0.04	—
LinearNoBias	N=256 / C=384	PyTorch (baseline)	0.04	—
LinearNoBias	N=512 / C=384	PyTorch (baseline)	0.04	—
LinearNoBias	N=1024 / C=384	PyTorch (baseline)	0.06	—
LinearNoBias	N=2048 / C=384	PyTorch (baseline)	0.06	—
MSAAttention	N=384 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.40	1.9x
MSAAttention	N=512 / C_msa=64 / C_pair=128 / H=8 / S=16	torch.compile	0.56	2.2x
MSAAttention	N=640 / C_msa=64 / C_pair=128 / H=8 / S=16	torch.compile	0.80	2.3x
MSAAttention	N=768 / C_msa=64 / C_pair=128 / H=8 / S=16	torch.compile	1.08	2.4x
OuterProductMean	N=384 / C_msa=64 / C_pair=128 / S=32	torch.compile	2.57	1.1x
OuterProductMean	N=512 / C_msa=64 / C_pair=128 / S=16	torch.compile	4.73	1.1x
OuterProductMean	N=640 / C_msa=64 / C_pair=128 / S=16	torch.compile	7.50	1.1x
OuterProductMean	N=768 / C_msa=64 / C_pair=128 / S=16	torch.compile	10.99	1.1x
SingleAttention	N=384 / C=384 / H=16	torch.compile	0.41	1.4x
SingleAttention	N=512 / C=384 / H=16	torch.compile	0.44	1.3x
SingleAttention	N=640 / C=384 / H=16	torch.compile	0.45	1.3x
SingleAttention	N=768 / C=384 / H=16	torch.compile	0.46	1.4x
SwiGLU (pair)	C=128 / M=147456	Triton fused	0.20	1.9x
SwiGLU (pair)	C=128 / M=262144	Triton fused	0.40	1.9x
SwiGLU (pair)	C=128 / M=409600	Triton fused	0.59	2.1x
SwiGLU (pair)	C=128 / M=589824	Triton fused	0.83	2.1x
SwiGLU (single)	C=384 / M=384	PyTorch (baseline)	0.04	—
SwiGLU (single)	C=384 / M=768	PyTorch (baseline)	0.04	—
Transition (MSA)	C=64 / M=8192	PyTorch (baseline)	0.15	—
Transition (MSA)	C=64 / M=12288	PyTorch (baseline)	0.19	—

Operation	Shape	Recommended	Time (ms)	Speedup
Transition (pair)	C=128 / M=147456	Triton fused	2.02	1.7x
Transition (pair)	C=128 / M=262144	torch.compile	4.02	1.5x
Transition (pair)	C=128 / M=409600	torch.compile	6.39	1.5x
Transition (pair)	C=128 / M=589824	Triton fused	9.20	1.5x
Transition (single)	C=384 / M=384	PyTorch (baseline)	0.13	—
Transition (single)	C=384 / M=768	PyTorch (baseline)	0.15	—
TriangleMulIncoming	N=384 / C=128	torch.compile	3.54	1.4x
TriangleMulIncoming	N=512 / C=128	Triton L3 (dual GEMM)	8.03	1.2x
TriangleMulIncoming	N=640 / C=128	Triton L3 (dual GEMM)	13.70	1.2x
TriangleMulIncoming	N=768 / C=128	compile (integrated)	18.44	1.2x
TriangleMulOutgoing	N=384 / C=128	compile (integrated)	3.29	1.6x
TriangleMulOutgoing	N=512 / C=128	compile (integrated)	7.42	1.3x
TriangleMulOutgoing	N=640 / C=128	torch.compile	12.61	1.3x
TriangleMulOutgoing	N=768 / C=128	torch.compile	16.60	1.3x